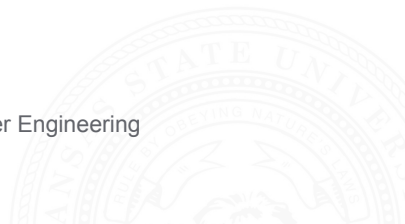


Single-Phase AC Circuits II

From voltage and current phasors to P , Q , $|S|$, and power factor

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Lecture outline

The concepts build in this order:

- 1 instantaneous power $p(t)$ and cycle-average real power P
- 2 reactive power Q , complex power S , and apparent power $|S|$
- 3 power factor, including leading or lagging, and
- 4 load/generator sign conventions and a complete worked example.

The numerical answer is incomplete until its sign, units, and physical meaning agree.

Starting problem

STARTING PROBLEM

**What current phasor follows from
 $V = 240\angle 0^\circ \text{ V}$ and $Z = 8 + j6 \Omega$?**



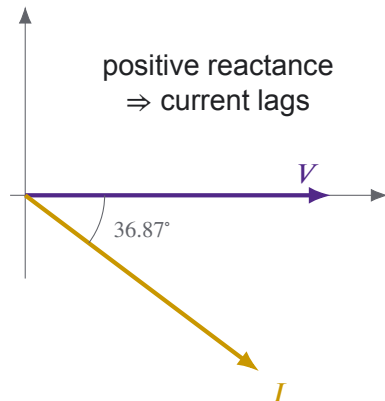
Starting-problem solution

1 Convert the impedance

$$8 + j6 = 10\angle 36.87^\circ \Omega.$$

2 Apply Ohm's law: $I = V/Z$

$$I = \frac{240\angle 0^\circ}{10\angle 36.87^\circ} = 24\angle (-36.87^\circ) \text{ A.}$$



Instantaneous power

Current enters the + terminal. Use RMS magnitudes with $\theta = \omega t$ and $\phi = \theta_V - \theta_I$.

$$v(t) = \sqrt{2}V_{\text{rms}} \cos \theta,$$

$$i(t) = \sqrt{2}I_{\text{rms}} \cos(\theta - \phi) = \sqrt{2}I_{\text{rms}} (\cos \theta \cos \phi + \sin \theta \sin \phi).$$

Using $2 \cos^2 \theta = 1 + \cos 2\theta$ and $2 \sin \theta \cos \theta = \sin 2\theta$,

$$\begin{aligned} p(t) = v(t)i(t) &= 2V_{\text{rms}}I_{\text{rms}} (\cos^2 \theta \cos \phi + \sin \theta \cos \theta \sin \phi) \\ &= V_{\text{rms}}I_{\text{rms}} [(1 + \cos 2\theta) \cos \phi + \sin 2\theta \sin \phi] \\ &= V_{\text{rms}}I_{\text{rms}} \cos \phi + \underbrace{V_{\text{rms}}I_{\text{rms}} (\cos 2\theta \cos \phi + \sin 2\theta \sin \phi)}_{\cos(2\theta - \phi)} \\ &= V_{\text{rms}}I_{\text{rms}} \cos \phi + V_{\text{rms}}I_{\text{rms}} \cos(2\theta - \phi). \end{aligned}$$

$p(t)$ is instantaneous power. It contains a constant term and a term that oscillates at 2ω .

Average real power P

The voltage and current period is $T = 2\pi/\omega$. Average $p(t)$ over this interval:

$$\begin{aligned} P &\equiv \frac{1}{T} \int_0^T p(t) dt \\ &= V_{\text{rms}} I_{\text{rms}} \cos \phi + V_{\text{rms}} I_{\text{rms}} \frac{1}{T} \int_0^T \cos(2\omega t - \phi) dt. \end{aligned}$$

The oscillating term has zero cycle average because

$$\begin{aligned} \frac{1}{T} \int_0^T \cos(2\omega t - \phi) dt &= \frac{[\sin(2\omega t - \phi)]_0^T}{2\omega T} \\ &= \frac{\sin(4\pi - \phi) - \sin(-\phi)}{4\pi} = 0. \end{aligned}$$

$$\boxed{P = V_{\text{rms}} I_{\text{rms}} \cos \phi.}$$

Capital P is cycle-average real power (W). Lowercase $p(t)$ is instantaneous.

Reactive power Q

Inductors and capacitors store energy in electric or magnetic fields and return it during another part of the cycle.

$$Q = V_{\text{rms}} I_{\text{rms}} \sin \phi \quad [\text{var}], \quad \phi = \theta_V - \theta_I.$$

This defines signed reactive power in sinusoidal steady state. With current referenced into the positive-voltage terminal:

- $\phi > 0$: current lags, $Q > 0$, inductive absorption.
- $\phi < 0$: current leads, $Q < 0$, capacitive supply.

For an ideal inductor or capacitor,

$$P = \frac{1}{T} \int_0^T p(t) dt = 0, \quad Q \text{ can be nonzero.}$$

P measures average net energy transfer. Q describes reactive exchange, which also requires current.

Recap: complex and apparent power

Use RMS phasors and the passive sign convention. The power angle is $\phi = \theta_V - \theta_I$.

$$\begin{aligned} S &= VI^* = |V||I|\angle\phi = |S|\angle\phi \\ &= |S|(\cos\phi + j\sin\phi) = P + jQ. \end{aligned}$$

$$P = |S| \cos\phi \text{ [W]}, \quad Q = |S| \sin\phi \text{ [var]}, \quad |S| = \sqrt{P^2 + Q^2} \text{ [VA]}.$$

Example: polar to rectangular

$$S = 5\angle 36.87^\circ \text{ kVA} \approx 4 + j3 \text{ kVA}.$$

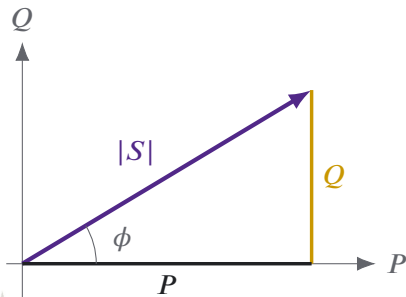
Thus $P = 4 \text{ kW}$, $Q = 3 \text{ kvar}$, and $|S| = 5 \text{ kVA}$.

Complex power S includes magnitude and angle. Apparent power $|S|$ is its magnitude.

Power triangle

Passive load convention: absorbed power is positive.

$$S = P + jQ, \quad |S|^2 = P^2 + Q^2.$$



$P > 0$: **real-power absorption**

Horizontal component.

$Q > 0$: **inductive**

Lagging current. Triangle up.

$Q < 0$: **capacitive**

Leading current. Triangle down.

Power factor

Passive load: $\text{pf} = P/|S| = \cos \phi$

Load type	Current vs. V	Load Q	Power-factor report
resistive	in phase	$Q = 0$	unity
inductive	lags	$Q > 0$	pf , lagging
capacitive	leads	$Q < 0$	pf , leading

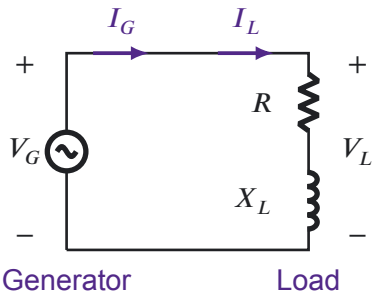
For fixed $P > 0$ and V_{rms} , with $\text{pf} > 0$,

$$P = V_{\text{rms}} I_{\text{rms}} \text{pf} \quad \Rightarrow \quad I_{\text{rms}} = \frac{P}{V_{\text{rms}} \text{pf}}.$$

For the same P and V , lower power factor means higher current, greater $I^2 R$ line losses, and more equipment capacity used.

Load and generator sign conventions

Subscripts: L = load, G = generator. Arrows define the current references.



An R - L load illustrates the references.

Load: power absorbed

I_L enters the + terminal.

$$S_L = V_L I_L^* = P_L + jQ_L.$$

$P_L > 0$: absorbs real power.

$Q_L > 0$: absorbs reactive power.

Generator: power delivered

I_G leaves the + terminal.

$$S_G = V_G I_G^* = P_G + jQ_G.$$

$P_G > 0$: supplies real power.

$Q_G > 0$: supplies reactive power.

For both P and Q : positive means absorption at the load and delivery at the generator.

Reactive power of loads and generators

Load: S_L is power absorbed. Generator: S_G is power delivered.

Reference	Positive reactive power	Negative reactive power
Load: Q_L	Absorbs reactive power Inductive behavior	Supplies reactive power Capacitive behavior
Generator: Q_G	Supplies reactive power	Absorbs reactive power

Generator example

$$P_G = 4 \text{ kW}, \quad Q_G = -1 \text{ kvar.}$$

Delivers 4 kW of real power and absorbs 1 kvar of reactive power.

A generator can supply real power while either supplying or absorbing reactive power.

SECTION

Worked example series-impedance calculation

Calculate current, complex power, and power factor.



Example setup

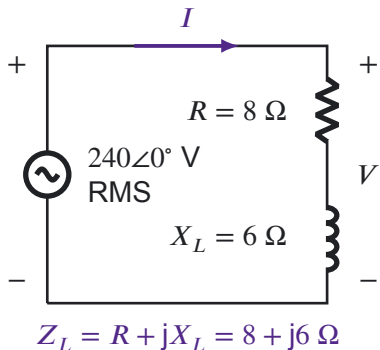
Given

$$V = 240\angle 0^\circ \text{ V}, \quad Z_L = 8 + j6 \text{ } \Omega.$$

V is an RMS phasor. Current enters the load's positive-voltage terminal.

Find

I , S_L , P_L , Q_L , $|S_L|$, and power factor.



series-impedance calculation Step 1

$$|Z_L| = \sqrt{8^2 + 6^2} = 10 \, \Omega$$

$$\angle Z_L = \arctan\left(\frac{6}{8}\right) = 36.87^\circ$$

$$Z_L = 10 \angle 36.87^\circ \, \Omega$$

Expected signs

- $X_L > 0$.
- current must lag.
- Q_L must be positive.



series-impedance calculation Step 2

$$I = \frac{V}{Z_L} = \frac{240\angle 0^\circ}{10\angle 36.87^\circ} = \boxed{24\angle(-36.87^\circ) \text{ A}}.$$

Phase Magnitude

24 A

240/10 has units $V/\Omega = A$.

Angle

-36.87°

Negative relative to V : current lags as predicted.

Form I with its original angle. Apply conjugation in the complex-power calculation.

series-impedance calculation Step 3

$$I^* = 24\angle(+36.87^\circ) \text{ A}$$

$$S_L = VI^* = (240\angle 0^\circ)(24\angle 36.87^\circ) = 5760\angle 36.87^\circ \text{ VA.}$$

$$S_L = 4608 + j3456 \text{ VA}$$



series-impedance calculation Step 4

Real power

$$P_L = 4.608 \text{ kW}$$

Positive: the load absorbs real power.

Reactive power

$$Q_L = +3.456 \text{ kvar}$$

Positive: the load is inductive.

Apparent power

$$|S_L| = 5.760 \text{ kVA}$$

$$VI = 240 \times 24.$$

Power factor

0.800 lagging

$$P/|S| = 0.8 \text{ and } Q > 0.$$

The six-word conclusion is: “inductive load, 0.800 power factor lagging.”

Complex-power summary

$$S = VI^* = P + jQ, \quad |S| = \sqrt{P^2 + Q^2}.$$

- **Power quantities:** P is cycle-average real power (W), Q is reactive power (var), and $|S|$ is apparent power (VA).
- **Power factor:** for a passive load, $\text{pf} = P/|S|$. Report the value together with leading or lagging.
- **Phase relationship:** lagging means current lags voltage ($Q_L > 0$). Leading means current leads voltage ($Q_L < 0$).
- **Sign convention:** for both P and Q , load absorption is positive and generator output is positive under the stated current references.